



Management and Conservation Article

Habitat Value of Intensively Established Pine Plantations for Northern Bobwhite

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ABSTRACT Declines in northern bobwhite (*Colinus virginianus*; bobwhite) populations in the southeastern United States may be partially attributable to loss of early successional plant cover associated with greater use of herbicides in forest management. We tested effects of 5 levels of operational plantation establishment intensity on vegetation communities and structure important for bobwhite in 1–5-year-old loblolly pine (*Pinus taeda*) plantations ($n = 4$) in the Outer Coastal Plain Mixed Forest of Mississippi. We compared results with values reported in the literature to calculate usable space for winter food, loafing, nesting, and brood-rearing at 2 levels of spatial resolution. Treatments ($k = 5$) reflected a range of management intensities and were combinations of mechanical site preparation, chemical site preparation (CSP), and herbaceous weed control (HWC). Coverage of winter food plants in the least intensive treatment was more than double that in the most intensive; however, differences in usable space of winter food cover were negligible due to improved accessibility in more intensive treatments. Although CSP reduced coverage of nonpine woody plants across all years, loafing cover reached adequate levels by year 3 in all treatments. Usable nesting cover was <4% across all years and treatments and was nearly eliminated by the reduction in herbaceous cover and visual screening cover following broadcast HWC. Optimal brood-rearing habitat was virtually absent in all treatments and years due to the lack of conjoint occurrence of bare ground and forb canopy. Although bobwhite habitat may have been promoted by formerly widespread plantation establishment methods that involved wide-scale soil disturbance, those established using newer methods with less soil disturbance are likely inadequate for most bobwhite habitat requirements. Efforts to provide bobwhite habitat in plantation-dominated landscapes may have to rely on management of thinned mid-rotation stands and permanent landscape features such as rights-of-way. The value of young plantations for bobwhite may be increased by reducing management intensity or increasing spacing between planting rows, thus increasing time before crown closure and providing opportunity for understory manipulations.

KEY WORDS *Colinus virginianus*, habitat quality, intensive plantation management, northern bobwhite, pine plantation, usable space.

As northern bobwhite (*Colinus virginianus*; hereafter bobwhite) populations have declined in the southeastern United States, some authors have attributed much of the cause to intensive plantation management practices and decreasing use of prescribed fire (Brennan 1991, Burger 2002). Use of prescribed fire has declined in the face of increasing costs and restricted use, and selective herbicide use has emerged as a silvicultural alternative (Haines et al. 2001, Wigley et al. 2002, Dubois et al. 2003). In 2002, approximately 433,000 ha of southern pine plantations received chemical site preparation (Dubois et al. 2003) and 286,000 ha received applications of herbaceous weed control (HWC; Shepard et al. 2004). However, increasing intensity of stand establishment can reduce abundance and diversity of woody and herbaceous plant species (Harrington and Edwards 1996, Miller et al. 1999, Schabenberger and Zedaker 1999, Jones et al. 2009). Given the widespread use of intensive forest management on large, pine-dominated parcels, it is imperative to understand the impacts of such management on vegetation communities vital to bobwhite.

Although several studies have documented effects of herbicide use during stand establishment on food plants important to bobwhite (McComb and Hurst 1987, Witt et al. 1993, Feken 1995, Keyser et al. 2003, Keyser and Ford 2006), most have been of short duration, and little information is available on impact of tank mixtures of herbicides, multiple herbicide applications, and combinations of mechanical and chemical site preparation (Miller

and Miller 2004). Options for improving bobwhite habitat have been explored in mature pine forests under intensive management (Jones and Chamberlain 2004, Welch et al. 2004, Burke et al. 2008). However, there is as yet no comprehensive assessment of the multiple habitat values potentially found in young, intensively established pine plantations.

Usable space has been defined as a point “associated with habitat compatible with the physical, behavioral, and physiological adaptations of bobwhites in a time-unlimited sense” (Guthery 1997:294). When habitat requirements are known, usable space can be calculated as the proportion of a given area compatible with those requirements. We assume this proportion can be used as a direct measure of habitat quality, with increasing habitat quality associated with greater proportions of usable space.

We investigated effects of 5 intensive establishment regimes on vegetation community structure and composition in loblolly pine (*Pinus taeda*) plantations during a 5-year period following site preparation and planting in southern Mississippi, USA. We compared the results with ranges and combinations of coverages reported in the scientific literature as consistent with bobwhite use or preference. Guthery et al. (2005) recommended evaluating temporal dynamics in usable space, because cover on a given area might change over time. Our goal was to determine when requirements for suitable winter foraging, loafing, nesting, and brood-rearing cover were met by each establishment regime during the successional dynamic period from plantation establishment to year 5.

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STUDY AREA

We studied bobwhite habitat on 4 loblolly pine plantations established on private commercially managed forest sites in George, Lamar, and Perry counties in the Coastal Flatwoods subregion of the Outer Coastal Plain Mixed Forest region in Mississippi (Bailey 1980). Elevations were 40–121 m above sea level; topography was gently undulating with site-specific elevational differences of 10–12 m (3 sites) to 27 m (1 site). Plantations were harvested during summer 2000–winter 2001 and averaged 66 ha. Common woody plants included sweetgum (*Liquidambar styraciflua*), various oaks (*Quercus* spp.), common persimmon (*Diospyros virginiana*), various hollies (*Ilex* spp.), eastern baccharis (*Baccharis halimifolia*), *Vaccinium* spp., wax myrtle (*Morella cerifera*), and American beautyberry (*Callicarpa americana*). Tungoil tree (*Vernicia fordii*) was common to 2 sites, which had previously been tung oil plantations. Common vines included lianas, such as poison ivy (*Toxicodendron radicans*), muscadine (*Vitis rotundifolia*), Virginia creeper (*Parthenocissus quinquefolia*), yellow jessamine (*Gelsemium sempervirens*), and various greenbriers (*Smilax* spp.), and early seral species such as sawtooth blackberry (*Rubus argutus*), southern dewberry (*R. trivialis*), and Japanese honeysuckle (*Lonicera japonica*). Herbaceous associates included threeawngrass (*Aristida* spp.), broomsedge (*Andropogon virginicus*), and various panic grasses (*Dicantheleum* spp.); forb communities were characterized by *Eupatorium* spp. and numerous asters (*Aster* spp., *Solidago* spp., and *Euthamia tenuifolia*). Two poorly drained plantations had a strong sedge (*Carex* spp. and *Cyperus* spp.) and rush (*Juncus* spp.) component. The climate was subtropical, with mean annual temperature of 18.7° C and mean annual rainfall of 159 cm (National Oceanic and Atmospheric Administration 2008). Soils of 2 plantations were completely composed of Susquehanna series, a somewhat poorly drained fine, smectitic, Vertic Paleudalf. Soils of a third plantation consisted entirely of McLaurin series, a deep, well-drained, coarse-loamy, siliceous, subactive, thermic Typic Paleudult. The fourth plantation was characterized by a mixture of McLaurin, Freestone (a very deep, moderately well-drained, fine-loamy, siliceous, semiactive, thermic Glossaquic Paleudalf), and Baxter (a very deep, well-drained, fine, mixed, semiactive, mesic Typic Paleudalf) soils.

METHODS

Study Design

Treatments were combinations of chemical site preparation (CSP), mechanical site preparation (MSP), and HWC, designed to reflect the range of operational intensities used on commercially managed forests of the southeastern United States. We applied CSP at all sites during July–August 2001 (yr 0) using a mixture of imazapyr (Chopper Emulsifiable Concentrate®, BASF Corp., Research Triangle Park, NC; 2.4 L/ha), glyphosate (Accord®, Dow AgroSciences LLC, Indianapolis, IN; 3.5 L/ha), and triclopyr (Garlon 4®, Dow AgroSciences LLC; 3.5 L/ha), and 1% volume to volume ratio of nonionic surfactant (Timberland 90®, UAP Timberland LLC, Monticello, AR) in a broadcast spray

Table 1. Treatments applied to establish loblolly pine plantations on 4 commercially managed forest sites in the Outer Coastal Plain Mixed Forest region of Mississippi, USA, to create a range of management intensities from low (M-B) to high (MC-Br2) for study of northern bobwhite habitat, 2002–2006.

Treatment ^a	Site preparation		Herbaceous weed control ^d	
	Mechanical ^b	Chemical ^c	Banded	Broadcast
M-B	×		Yr 1	
C-B		×	Yr 1	
MC-B	×	×	Yr 1	
MC-Br	×	×		Yr 1
MC-Br2	×	×		Yr 1 and 2

^a Treatments are abbreviated to indicate type of site preparation followed by type of herbaceous weed control. Thus M-B = mechanical site preparation and banded herbaceous weed control; C-B = chemical site preparation and banded herbaceous weed control; MC-B = mechanical and chemical site preparation and banded herbaceous weed control; MC-Br = mechanical and chemical site preparation and broadcast herbaceous weed control; MC-Br2 = mechanical and chemical site preparation and 2 yr of broadcast herbaceous weed control.

^b Single pass with a combination plow to subsoil, disk, and bed.

^c Tank mix of imazapyr (Chopper Emulsifiable Concentrate®, BASF Corp., Research Triangle Park, NC; 2.4 L/ha), glyphosate (Accord®, Dow AgroSciences LLC, Indianapolis, IN; 3.5 L/ha), and triclopyr (Garlon 4®, Dow AgroSciences LLC; 3.5 L/ha).

^d Sulfometuron methyl and hexazinone (Oustar®, E. I. du Pont de Nemours and Company, Inc., Wilmington, DE; 0.9 kg/ha). Banded treatments covered the width of the planting bed (1.5 m), yielding 50% coverage.

solution of 93.6 L/ha. We applied MSP (subsoiling, disking, and bedding) during September–December 2001 (yr 0) using one pass of a combination plow, which yielded alternating 1.5-m-wide strips of plowed and unplowed ground. We applied HWC consisting of sulfometuron methyl and hexazinone (Oustar®, E. I. du Pont de Nemours and Company, Inc., Wilmington, DE; 0.9 kg/ha) during March–April 2002 (yr 1) and March–May 2003 (yr 2); we applied HWC either in 1.5-m bands over the tops of pine seedlings, resulting in 50% total coverage, or broadcast it aerially over the entire experimental unit.

Treatments from least to most intensive were MSP and 1 year banded HWC (M-B); CSP and 1 year banded HWC (C-B); both CSP and MSP with 1 year banded HWC (CM-B); both CSP and MSP with 1 year broadcast HWC (CM-Br); both CSP and MSP with 2 years broadcast HWC (CM-Br2; Table 1). We randomly assigned each treatment to an area ≥ 8 ha within each plantation ($n = 4$), creating a randomized complete block design. Apart from these treatments, we standardized management across all plots. We planted loblolly pines on each plantation during the winter of 2001–2002 using 3.0×2.1 -m spacing (1,551 trees/ha); each participating company used its own seedlings. Two plantations were machine-planted, and 2 were hand-planted due to high coarse woody debris loads. We fertilized all plantations in April 2002 with a broadcast application of di-ammonium phosphate at 280 kg/ha.

Sampling

We measured continuous coverage of ferns, forbs, graminoids, vines, woody plants, litter, and bare ground along 10 randomly placed and oriented 30-m line transects in each

experimental unit during June 2002–2006. Multiple layering of plants often resulted in total coverage >100%. We tabulated bare ground and debris only when there was ≥ 50 cm from ground level to any vegetation canopy. In conjunction with each transect, we measured visual obstruction using a 1.8-m Nudds density board (Nudds 1977) stratified into 6 0.3-m sections. We viewed the board at a distance of 10 m in each cardinal direction from a height of 1.5 m and estimated obstruction of each section in 20% increments. The observer took readings along each transect at meters 7 and 22, yielding 8 observations associated with each transect. We analyzed estimates using the mid-point of each range as the estimated coverage, yielding maximum obstruction of 90%.

We reviewed the literature for studies of bobwhite habitat use and selected ranges of vegetation coverage for winter food, loafing, nesting, and brood-rearing cover. Schroeder (1985) considered winter food cover to be optimized across the range of 25–75% canopy coverage of preferred bobwhite herbaceous food plants combined with 30–60% coverage of bare ground or light litter. Assessments of this model have led to proposed modifications, including reducing minimum optimal food-plant coverage to 12.5% (Tonkovich and Stauffer 1993) and increasing the range of bare ground–light litter coverage to 50–60% (O’Neil 1993) or 65–82% (Tonkovich and Stauffer 1993). We therefore expanded the range of acceptable plant food coverage to 12.5–75% and that of bare ground–light litter coverage to 30–82%. Following the methodology of Landers and Johnson (1976), Brazil (1993) assigned importance values (IV) of 1–16 to winter food-plant species based on percentage occurrence and percentage volume in diet of bobwhites in Mississippi; we used species with $IV \geq 4$ to represent preferred species. Optimal loafing cover has been described as having 40–80% coverage of woody vegetation ≤ 2 m in height and preferably in patches ≥ 188 m² in area (Schroeder 1985). Schroeder (1985) summarized optimal nesting cover as 40–60% coverage of herbaceous plants, 40–60% of which are grasses, with canopy height of ≥ 40 cm; subsequent studies have found cover and visual obstruction within these ranges (Townsend et al. 2001, Lusk et al. 2006). Visual obstruction requirements for brood-rearing habitat may require canopy heights approaching 60 cm (Taylor and Burger 2000). Bare ground is important to allow accessibility by bobwhite chicks that might become exhausted in thick vegetation or debris (Hurst 1972). Reports of bare ground coverage at brooding sites range from 14% to 25% (Speake and Sermons 1986, Burger et al. 1994, Taylor et al. 1999, Taylor and Burger 2000, Carver et al. 2001). Taylor et al. (1999) reported that forb coverage was also important in selection of diurnal brood-rearing sites; observed forb coverages ranged from 34% to 47% (Burger et al. 1994, Taylor et al. 1999, Taylor and Burger 2000).

We used these combinations of coverage and visual obstruction to represent usable space and evaluated habitat at 2 levels of spatial resolution. Because visual obstruction measurements applied only to the transect level, we calculated both a transect-level and a meter-level value for usable space for loafing, nesting, and brood-rearing cover,

with the meter-level value based only on coverage of vegetation, bare ground, and debris. We considered individual meters along each transect as a sampling point and calculated percentage of meters ($n = 300$) within each experimental unit that met requirements for each cover type. At the transect level, we calculated mean obstruction height ($n = 8$) and coverage of vegetation types, bare ground, and debris to determine usable space. Winter food cover did not require an obstruction reading, so we calculated usable space at the meter level only. Bobwhite in Kansas (USA) appeared to select nesting sites based on site, rather than patch, characteristics (Taylor et al. 1999); thus, we considered meter-level resolution more appropriate than transect-level for gauging usable space.

Data Analysis

We used a repeated-measures mixed-model analysis of variance to test for main effects of year, treatment, and year $\times$ treatment interaction for coverage of important winter food plants; coverage of bare ground, debris, forbs, graminoids, herbaceous plants, and woody plants; and coverage of each Nudds board section. We compared means among treatments ($n = 5$) and years ($n = 5$) in SAS PROC MIXED (SAS Institute, Cary, NC). We treated plantations (i.e., blocks, $n = 4$) as the random effect, years as the repeated effect, and treatment $\times$ plantation as the subject. For each analysis, we selected the combination of covariance structure and use of the random statement that minimized Akaike’s Information Criterion corrected for small sample size (AIC_c ; Littell et al. 2006, Gutzwiller and Riffell 2007). We used the Kenward–Roger correction in denominator degrees of freedom for repeated measures to avoid inflated Type I error (Littell et al. 2006, Gutzwiller and Riffell 2007). We considered differences significant if $P \leq 0.05$. We used LSMEANS SLICE to identify a treatment effect within years following a significant interaction (Littell et al. 2006). We compared means using Fisher’s least significant difference with the LSMEANS PDIF option (Littell et al. 2006). For ease of data interpretation, we present actual means although we conducted analyses of vegetation coverage, bare ground, and debris on transformed data to improve homogeneity of variance and normality. For winter food, loafing, nesting, and brood-rearing cover, we calculated mean percentage usable space and a 95% confidence interval.

RESULTS

Vegetation Response

Woody coverage in M-B averaged 8.2% greater than other treatments across all 5 years (Table 2). Woody coverage increased on all treatments across years, reaching 67–82% by year 5. Coverage of forbs, graminoids, herbaceous plants, debris, and bare ground all exhibited significant year $\times$ treatment interactions.

Relative to banded HWC, broadcast HWC reduced herbaceous coverage and its primary components of forbs and graminoids in the years we applied it (Table 2). Coverage in broadcast HWC treatments increased the year following application to levels equivalent with treatments receiving

Table 2. Percent coverage of ground cover types important to northern bobwhite in 1–5-year-old loblolly pine plantations under 5 regimens of establishment intensity varying from low (M-B)^a to high (CM-Br2) in the Outer Coastal Plain Mixed Forest region of Mississippi, USA, 2002–2006.^b

Ground cover	Yr	Treatment (Trt)										<i>P</i> -values ^c		
		M-B		C-B		CM-B		CM-Br		CM-Br2		Yr	Trt	Yr × trt
		$\bar{x}$	SE	$\bar{x}$	SE	$\bar{x}$	SE	$\bar{x}$	SE	$\bar{x}$	SE			
Forbs	1	12 Aab ^d	2	19 Aa	9	11 Aa	3	2 a	0	1 Ba	1	≤0.001	0.025	≤0.001
	2	23 Ac	7	28 Ab	10	27 Ab	9	28 b	5	7 Ba	5			
	3	14 bc	4	20 ab	8	24 b	7	29 b	9	14 bc	7			
	4	9 ab	2	14 a	6	16 ab	3	14 c	4	12 c	3			
	5	6 a	2	7 c	4	11 a	3	10 c	3	6 ab	2			
Graminoids	1	15 Aa	3	18 Aa	6	9 Aa	2	2 Ba	1	1 Ba	1	≤0.001	0.022	0.009
	2	29 ABbc	7	33 Ab	5	22 ABbc	6	21 BCb	7	10 Cb	3			
	3	38 c	6	47 c	9	45 d	8	43 c	7	50 c	11			
	4	28 c	3	37 bc	8	34 cd	5	36 c	6	22 d	1			
	5	17 ab	2	15 a	1	18 ab	2	15 b	2	17 b	3			
Debris	1	44 Aa	4	59 ABa	13	59 Ba	4	71 Ba	7	72 Ba	6	≤0.001	≤0.001	≤0.001
	2	13 Ab	4	22 Bb	4	15 ABb	7	20 Bb	3	66 Ca	9			
	3	17 bc	2	21 b	1	17 bc	2	13 b	2	19 b	3			
	4	13 Abc	3	18 ABb	3	16 Ab	2	19 Bb	3	28 Bb	3			
	5	21 Ac	2	36 BCc	3	26 ABc	2	34 Cc	3	44 Cc	4			
Herbaceous ^e	1	27 ABab	4	37 Aa	15	20 Ba	3	3 Ca	1	3 a	1	≤0.001	≤0.001	≤0.001
	2	51 Acd	1	60 Abc	5	49 Ab	8	49 Ab	9	17 Bb	5			
	3	53 c	6	68 c	6	70 c	14	73 c	8	64 c	12			
	4	38 bd	4	52 b	3	52 b	5	50 b	4	37 d	4			
	5	23 a	4	23 a	3	30 a	3	26 d	3	18 b	3			
Woody ^f	1	7	1	4	2	3	0	2	1	2	1	≤0.001	0.014	0.286
	2	22	3	15	2	15	4	16	3	8	1			
	3	48	10	34	7	30	5	40	4	37	5			
	4	59	4	47	8	45	4	53	7	65	8			
	5	82	4	72	10	67	6	79	12	76	5			
Bare ground	All	44 A	7	34 B	6	32 B	6	38 B	7	38	7	≤0.001	0.048	≤0.001
	1	15 Aa	2	2 Ba	0	14 Aa	3	22 Aa	8	22 Aa	7			
	2	1 Ab	1	0 Ab	0	2 ABb	1	6 b	3	7 bc	4			
	3	1 b	0	0 b	0	1 b	1	2 b	1	8 b	3			
	4	2 b	0	2 a	0	2 b	1	2 b	1	4 bc	1			
	5	1 b	0	1 c	0	1 b	0	1 b	0	1 c	0			

^a See Table 1 for treatment descriptions.^b Actual means presented; we performed analysis on square-root-transformed data. *P*-values and treatment differences correspond to least-square means.^c Df were as follows: yr_{Forb} = 4,40; trt_{Forb} = 4,5,45; yr × trt_{Forb} = 16,30,8; yr_{Graminoid} = 4,56,4; trt_{Graminoid} = 4,16,6; yr × trt_{Graminoid} = 16,57,2; yr_{Debris} = 4,54,8; trt_{Debris} = 4,18,3; yr × trt_{Debris} = 16,55,8; yr_{Herbaceous} = 4,53,4; trt_{Herbaceous} = 4,15; yr × trt_{Herbaceous} = 16,54,5; yr_{Woody} = 4,56,6; trt_{Woody} = 4,19,1; yr × trt_{Woody} = 16,57,3; trt_{Bare ground} = 4,44,6; trt_{Bare ground} = 4,3,95; yr × trt_{Bare ground} = 16,28,3.^d Within-yr treatment means followed by the same uppercase letter do not differ significantly ($\alpha = 0.05$). Within each ground-cover class, within-treatment means followed by the same lowercase letter do not differ significantly ($\alpha = 0.05$).^e Includes ferns, forbs, and graminoids.^f All yr differed ($P \leq 0.001$). tr 1: $\bar{x} = 4$, SE = 1; yr 2: $\bar{x} = 15$, SE = 1; yr 3: $\bar{x} = 38$, SE = 3; yr 4: $\bar{x} = 54$, SE = 3; yr 5: $\bar{x} = 75$, SE = 3.

banded HWC. Herbaceous coverage peaked in year 3, averaging 66%, then declined to 24% in year 5. Graminoid coverage responded similarly, averaging 45% at its peak in year 3, and then declining to 16% in year 5. The year 2 HWC application in CM-Br2 delayed peak forb production by 1 year relative to treatments with 1 HWC application.

Mechanical site preparation increased bare ground coverage in year 1 by 7 times over CSP alone (Table 2). Treatments with broadcast HWC maintained greater bare-ground levels than those with banded HWC for 1 year following the last application. Bare ground decreased quickly in treatments receiving banded HWC, falling from an average of 10% in year 1 to 1% in years 2–5. Debris coverage was more complex. In year 1, greater intensity treatment produced 67% debris coverage in CM-B, CM-Br, and CM-Br2 compared with 44% in M-B. In year 2, broadcast HWC maintained greater debris levels in CM-Br2, whereas coverage in remaining treatments decreased by

an average of 41% to 17%. All treatments were equivalent in year 3, the same year that herbaceous cover peaked across all treatments. Debris coverage was again greater in CM-Br2 in years 4 and 5, this time due to increasing pine coverage, which shaded out competitors and increased the area covered by pine litter.

Coverage of winter food plants decreased with increasing management intensity (Table 3). Across all years, coverage averaged 50% in M-B, 35% in the intermediate treatments, and 18% in CM-Br2. Food-plant coverage increased from a treatment mean of 8% in year 1 to a peak of 52% in year 3, then declined to 37% by year 5.

Visual obstruction increased across years at all heights in all treatments, reaching $\geq 76\%$ for all board sections by year 5 (Table 4). Obstruction in the lowest section (0.0–0.3 m above ground) decreased with increasing management intensity across all years, although all means approached maximum obstruction by year 3 (84–89%). Year 1

Table 3. Percent coverage of important winter northern bobwhite food plants in 1–5-year-old loblolly pine plantations under 5 regimens of establishment intensity varying from low (M-B)^a to high (CM-Br2) in the Outer Coastal Plain Mixed Forest region of Mississippi, USA, 2002–2006.^b

Yr	Treatment (Trt)					P-values ^c		
	M-B	C-B	CM-B	CM-Br	CM-Br2	Yr	Trt	Yr × trt
1								
$\bar{x}$	16	8	10	2	2	<0.001 ^d	<0.006	0.248
SE	2	3	3	1	0			
2								
$\bar{x}$	54	31	50	34	5			
SE	7	9	16	10	3			
3								
$\bar{x}$	75	52	59	54	24			
SE	11	12	15	9	8			
4								
$\bar{x}$	58	44	45	34	32			
SE	3	9	7	5	4			
5								
$\bar{x}$	48	40	37	33	29			
SE	5	4	2	4	3			
All								
$\bar{x}$	50 A ^e	35 B	40 AB	31 BC	18 C			
SE	5	5	6	5	3			

^a See Table 1 for treatment descriptions.^b Actual means presented; we performed analysis on log-transformed data. P-values and treatment differences correspond to least-square means.^c Df were 4,12 for yr; 4,13.8 for treatment; and 16,17.9 for interaction.^d Yr followed by the same upper case letter are not significantly different ($\alpha = 0.05$): 1 A, 2 B, 3 C, 4 D, 5 BD.^e Treatment means followed by the same upper case letter are not significantly different ($\alpha = 0.05$).

differences in the 0.3–0.6-m section reflected the cumulative impacts of combined site preparation and broadcast HWC; by year 2, broadcast HWC was the only consistent factor reducing obstruction in this section. In the 0.6–0.9-m section, year 1 treatment differences were not associated with particular treatment elements; however, the second broadcast HWC application reduced obstruction in CM-Br2 relative to other treatments in year 2. Obstruction from 0.9 m to 1.5 m was unaffected by treatment. Differences in the 1.5–1.8-m section did not reflect management intensity, with the least and most intensive treatments similar.

Usable Space

Usable winter food cover was created on $\leq 29\%$ ($\bar{x} = 13.2\%$) of meters across all years and treatments (Fig. 1). Although C-B appeared to be superior to other treatments, this was due to response at one site, and the 95% confidence interval for C-B included zero in all years. With the exception of year 1, 95% confidence intervals overlapped among all treatments, and 13 of 25 95% confidence intervals included zero. Loafing cover at the meter level increased steadily across all treatments from year 1 to year 4 with little difference among treatments, mirroring the growth of woody vegetation. At the transect level, there was little difference between years 1 and 2, followed by a strong increase into years 3 and 4. Nesting cover was usable on $<4\%$ ($\bar{x} = 2\%$) of all meters sampled in each year × treatment combination, with the 95% confidence interval including zero for 14 of 25 means. At the transect level, means were marginally greater ($\bar{x} = 4\%$), but all 95% confidence intervals included zero. Usable brood-rearing cover at the meter level averaged $<1\%$ across all year ×

treatment combinations, and all 95% confidence intervals included zero; at the transect level, only one transect among the 1,000 we surveyed met criteria for usability.

DISCUSSION

Historically, regenerating and newly established pine plantations in the southeastern United States have provided 2–4 years of early successional habitat suitable for bobwhites (Schultz and Wilhite 1974, Burger 2001). Mechanical site preparation during the 1960s and 1970s often included raking and windrowing followed by broadcast disking and burning (Fox et al. 2007), resulting in a canopy of forbs with abundant bare ground suitable for winter foraging and brood-rearing (Burger 2001). As succession proceeded into grassy and shrub stages, winter food and brood-rearing cover deteriorated, whereas loafing and nesting cover improved until eliminated by crown closure. Plantations in today's intensively managed pine forests are often managed on ≤ 30 -year rotations; precrown closure plantations may compose 15–25% of such landscapes, making the utility of young plantations especially important. Maximizing usable space in young plantations necessitates providing cover sufficient to meet the needs of as many life processes as possible within the stand.

When examining usable space in each treatment, it is important to remember that results are based on optimal cover, and that somewhat less suitable yet potentially usable cover may also be present. Also, optimal cover values are taken primarily from observational studies of space use, and may not represent true optimums as measured by fitness benefits. The point concept of usable space as articulated by

Table 4. Visual obstruction (%) of Nudds density board sections in 1–5-year-old loblolly pine plantations under 5 regimens of establishment intensity varying from low (M-B)^a to high (CM-Br2) in the Outer Coastal Plain Mixed Forest region of Mississippi, USA, to evaluate habitat for northern bobwhite, 2002–2006.^b

Section ht (m)	Yr	Treatment (trt)										P-values ^c		
		M-B		C-B		CM-B		CM-Br		CM-Br2		Yr	Trt ^c	Yr × trt
		$\bar{x}$	SE	$\bar{x}$	SE	$\bar{x}$	SE	$\bar{x}$	SE	$\bar{x}$	SE			
0.0–0.3	1	65	2	61	3.8	48	3	32	2	31	2	≤0.001	0.020	0.093
	2	84	2	75	2.2	80	2	75	2	53	3			
	3	89	0	84	1.4	85	1	86	1	85	1			
	4	90	0	88	1.0	89	0	89	1	89	0			
	5	89	0	89	0.5	89	1	88	1	85	1			
0.3–0.6	All	A		AC		AB		BC		C		≤0.001	0.002	≤0.001
	1	38 A	3	36 A	4.3	20 B	2	10 C	2	10 C	2			
	2	78 A	2	65 B	3.1	71 AB	2	65 B	3	44 C	3			
	3	88	1	80	1.8	81	2	82	2	80	2			
	4	90		86	1.5	87	1	87	1	88	1			
0.6–0.9	5	88	1	87	0.7	87	1	84	1	81	2	≤0.001	0.045	0.004
	1	13 AB	2	19 A	3.5	5 BC	1	2 C	1	4 BC	1			
	2	65 A	3	54 B	3.6	58 AB	3	55 AB	3	37 C	3			
	3	84	1	74	2.2	76	2	79	2	73	3			
	4	90	0	83	2.0	86	1	86	1	87	1			
0.9–1.2	5	86	1	84	1.0	83	1	81	2	78	2	≤0.001	0.134	0.324
	1	4	1	9	2.2	2	1	1	0	2	1			
	2	51	4	41	3.4	45	3	42	3	30	2			
	3	79	2	66	2.7	69	2	73	3	67	3			
	4	88	1	81	2.2	85	1	85	1	85	1			
1.2–1.5	5	86	1	82	1.2	81	1	78	2	78	2	≤0.001	0.062	0.738
	1	2	1	5	1.7	1	0	0	0	2	1			
	2	41	3	29	2.8	33	3	33	3	25	2			
	3	74	2	59	3.1	62	3	67	3	62	4			
	4	87	1	78	2.7	83	2	82	1	83	2			
1.5–1.8	5	85	1	80	1.6	79	2	76	2	76	3	≤0.001	0.036	0.782
	1	1	1	3	1.5	1	0	0	0	1	1			
	2	33	3	20	2.1	26	3	27	2	21	2			
	3	69	3	51	3.5	55	3	63	4	60	4			
	4	85	1	76	2.9	82	2	82	2	82	2			
All	5	85	1	80	1.7	78	2	76	2	76	2			
	All	55 A	8	45 B	7.0	49 B	8	51 AB	8	50 AB	8			

^a See Table 1 for treatment descriptions.

^b Actual means presented; we performed analysis on log-transformed data. *P*-values and treatment differences correspond to least-square means.

^c Df were as follows: yr_{0.0–0.3} = 4,15.9; trt_{0.0–0.3} = 4,4.73; yr × trt_{0.0–0.3} = 16,6.26; yr_{0.3–0.6} = 4,53.6; trt_{0.3–0.6} = 4,14.5; yr × trt_{0.3–0.6} = 16,54.3; yr_{0.6–0.9} = 4,60; trt_{0.6–0.9} = 4,12; yr × trt_{0.6–0.9} = 16,60; yr_{0.9–1.2} = 4,60; trt_{0.9–1.2} = 4,12; yr × trt_{0.9–1.2} = 16,60; yr_{1.2–1.5} = 4,58.6; trt_{1.2–1.5} = 4,23.8; yr × trt_{1.2–1.5} = 16,59.2; yr_{1.5–1.8} = 4,58.3; trt_{1.5–1.8} = 4,24.3; yr × trt_{1.5–1.8} = 16,59.

Guthery (1997) is binary, rather than continuous, and cover values that fall outside the accepted range are considered unusable. By including results from several studies of bobwhite habitat use, we attempted to ensure an appropriately inclusive range of cover values. In addition, calculating usable space at spatial grains meaningful to bobwhite may present results that are contrary to expectations based on stand-level measurements. The reduced resolution of broader scale measurements may obscure the lack of conjoint occurrence of all necessary habitat characteristics at finer resolutions important to bobwhite.

Although food may not be a limiting factor for most bobwhite populations (Guthery 2002), important food plants may be characteristic of plant communities that provide habitat structure and species composition to which bobwhites are adapted. Bobwhite densities in regenerating pine plantations have been correlated with abundance of important food plants (Brunswick and Johnson 1972). Previous researchers also have observed a correlation between decreasing use of young pine stands by bobwhite

and increasing vegetation thickness (Brunswick and Johnson 1972, Sweeney et al. 1981). This relationship was acknowledged by Schroeder's (1985) combination of bare ground–light litter and food-plant coverage requirements for optimal winter food habitat suitability. Low food-plant coverage following herbaceous weed control limited usable winter food space in year 1, and lower than optimal litter coverage began limiting usability the year following the last herbaceous weed control application, even as food-plant coverage increased.

Mechanically site-prepared pine plantations provide pulses of bobwhite food plants 1–3 years following treatment (Brunswick and Johnson 1972, Sweeney et al. 1981, Felix et al. 1986, Witt et al. 1993). Chemical treatment of plantations targeting woody competitors can enhance growth of herbaceous plants (Knowe et al. 1992, Miller et al. 1995, Mihalco 2004) and promote species (e.g., legumes) with known resistance to particular herbicides during site preparation or release operations (Hurst 1987, Witt et al. 1993, Shaw et al. 2001). We formulated our site-preparation

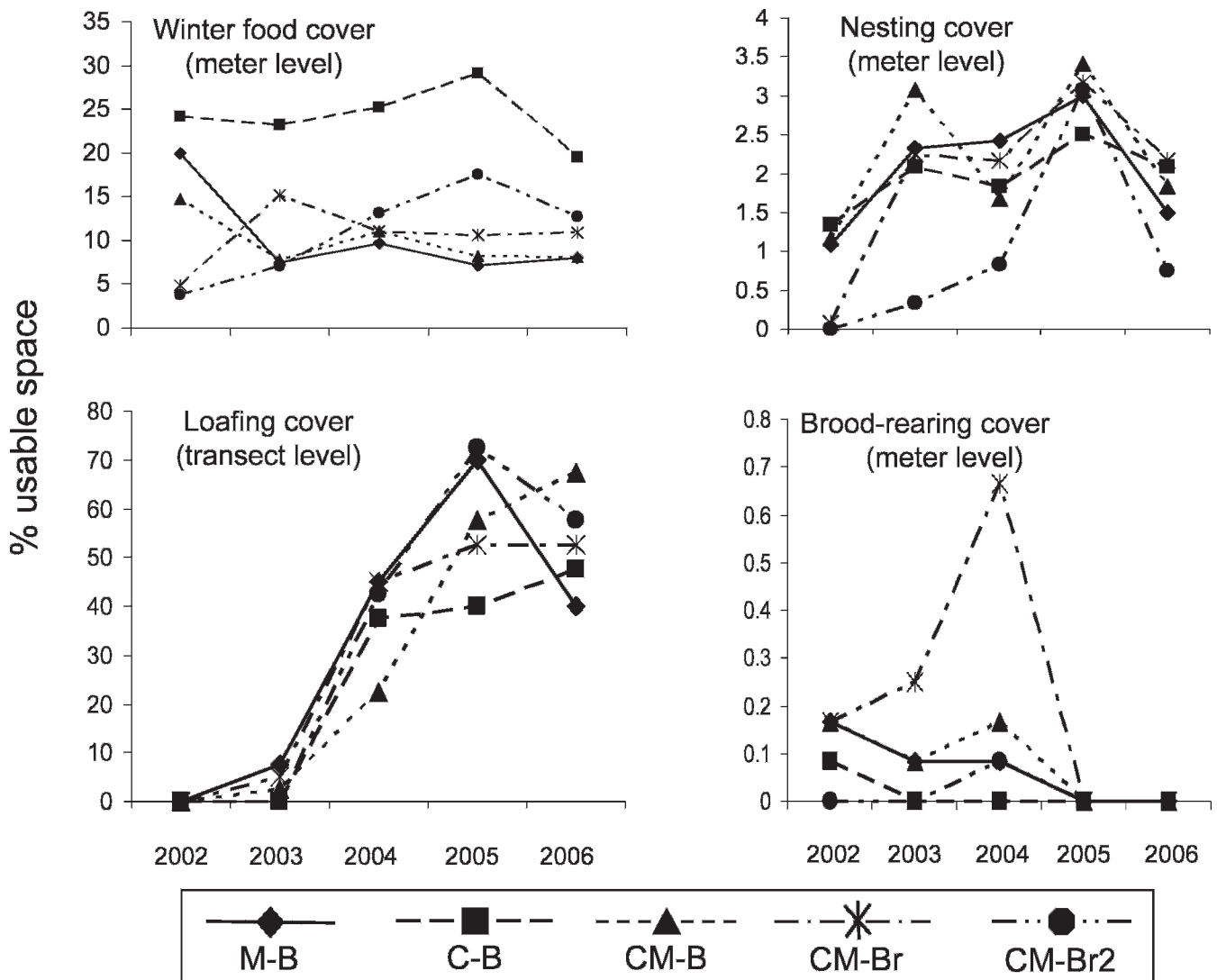


Figure 1. Percent usable space for habitat variables important to northern bobwhite in intensively established loblolly pine plantations 1–5 years postestablishment in southern Mississippi, USA, 2002–2006. Resolution level (transect or meter) reflects the need for accessibility (winter food, loafing, and brood-rearing cover) or assumed selectivity (nesting cover) of bobwhite for cover at that scale. Treatments consisted of combinations of mechanical site preparation (M), chemical site preparation (C), and banded (B) or broadcast (Br) herbaceous weed control applied for 1 year or 2 years after planting.

herbicide mix to provide broad-spectrum control of woody plants, opening the way for a strong herbaceous response. This potential asset was limited by broadcast herbaceous-weed control, which greatly reduced herbaceous plant coverage during the year in which we applied it and likely was similarly impacted by banded herbaceous weed control along planting beds (Mihalco 2004).

Loafing cover need not be ubiquitous; rather, loafing cover should be scattered throughout an area so that it is readily available from any point within the habitat. Kassinis and Guthery (1996) recommended that escape cover be located ≤ 100 m from any given point to be compatible with bobwhite flight behavior. Therefore, we believe it likely that any of these treatments provided adequate loafing cover beginning in year 3, when $\geq 20\%$ of transects provided optimal coverage (Fig. 1). Mechanical site preparation typically suppresses noncrop woody vegetation for 1–2 years (Stransky et al. 1986, Miller et al. 1995, Lauer and Zutter 2001), whereas chemical treatment may substantially control

noncrop woody cover for ≥ 10 years (Quicke et al. 1996, Miller et al. 1999). In our study, woody plant coverage (including planted pines) was greater in the mechanically prepared treatment. Pine trees were the primary source of woody coverage in more intensive treatments due to more complete competition control and consequent superior growth rate. Although pine height was >2.0 m by year 3, self-pruning of lower limbs had not occurred at this time. Furthermore, pines in the 3 more intensive treatments closed ranks along rows by year 4, providing access to long swaths of cover with low ceilings. Therefore, pines at these ages may have provided adequate loafing cover for bobwhite. However, the lack of loafing cover in years 1 and 2 may have reduced the value of plantations for other activities.

Although coverage of vegetation types affiliated with nesting cover appeared to be reasonably close to optimum requirements at the broader scale of the experimental unit (≥ 8 ha), these ratios were seldom found at the 1-m scale. The low incidence of usable space indicates that these

plantations were of low quality as nesting habitat regardless of treatment. Although broadcast herbaceous weed control appeared to more greatly reduce nesting cover during the year we applied it than did banded herbaceous weed control, it is unlikely that adequate nesting cover would develop during year 1 in any plantation regardless of whether or what type of herbaceous weed control was applied. In addition, although 2 years of broadcast herbaceous weed control promoted faster dominance of pine trees and delayed development of adequate visual obstruction relative to other treatments, all 95% confidence intervals overlapped in years 2–5, indicating that this made little if any difference.

Brood-rearing habitat was virtually precluded due to lack of conjoint occurrence of bare ground and forb coverage requirements. Mechanical site preparation did create some instances of adequate bare ground in year 1, primarily within the strips created by the combination plow; however, herbaceous weed control applications in spring of year 1 prevented development of adequate forb cover. Chemical site preparation rarely provided sufficient bare ground, similar to studies in 2-year-old loblolly pine plantations released with imazapyr in South Carolina (USA; Feken 1995) and in tall fescue fields treated with herbicide (Gruchy 2007), indicating that herbicide may, in general, require supplementation with mechanical disturbance to provide brood-rearing habitat. Burke et al. (2008) found that one prescribed burn following herbicide treatment did not create adequate bare ground for brood-rearing bobwhite in mature pine stands, indicating the necessity for repeated burns to restore and maintain such habitat following years of litter build-up.

Given the low quality of habitat found in these intensively established plantations, management of permanent landscape features in close juxtaposition to open-canopied pine stands will likely be necessary to provide the full complement of bobwhite habitat in areas dominated by intensive pine silviculture. Mid-rotation and mature stands may offer opportunities to manage ground-story vegetation with herbicide and fire (Jones and Chamberlain 2004, Welch et al. 2004), particularly after thinning (Grelen and Enghardt 1973, Cram et al. 2002). However, most thinning regimes on commercial timberland do not result in long-term open-canopied conditions. Managing permanent landscape features such as rights-of-way with rotational strip disking may improve structural characteristics for nesting and brooding bobwhite and provide vital habitat components (Greenfield et al. 2003, 2005). Timing and distribution of regeneration cuts may be critical to providing available early succession areas for bobwhite in areas dominated by even-aged management (Sweeney et al. 1981). Landscape-scale management that ensures interspersed early seral communities among young plantations, recently thinned stands (White et al. 2005), and permanent landscape features will be necessary to provide adequate habitat for bobwhites.

The silvicultural methods we used are common throughout the Outer Coastal Plain Mixed Forest, and we assume applicability of our results to this region. However, managed pine systems in other regions of the southern United States have different plant assemblages that may require different

site preparation prescriptions, potentially producing different outcomes for bobwhite. However, regardless of location, the goal of herbicide application is to diminish or eliminate particular vegetation complexes (woody or herbaceous) during specific portions of the stand establishment period, within targeted spatial extents (i.e., within the row [banded] or within the stand [broadcast]). It is this vegetation control that dramatically affects usable space for bobwhite. Consequently, one might expect similar patterns of uniformly low usable space in other regions, but we did not test this. Bobwhites were present on each of our study sites, but bobwhite use of plantations during both breeding season and winter was low and exhibited no year or treatment effects (Hanberry 2007). Because our experimental units were smaller than the typical bobwhite home range we did not conduct any work on demographic parameters. However, our study frames clear mechanistic hypotheses about habitat deficiencies in industrially managed pine plantations that can and should be tested with additional studies.

In areas dominated by intensive pine management, the overall lack of open canopy may be the greatest barrier to providing suitable habitat conditions for bobwhite. One goal of intensive pine plantation establishment is to shorten the open-canopied phase and allow crop trees to shade out inter-specific competitors. Increasing the spacing of planted pines can be a financially acceptable alternative under certain conditions (Huang and Kronrad 2004, VanderSchaaf and South 2004), increasing the length of time newly established plantations maintain an open canopy and allowing opportunities for understory manipulations (Radtke and Burkhart 1999). Periodic disking of interspersed blocks or strips of between-row vegetation could provide a patchwork of mixed early seral stages targeting different life requirements, increasing and extending the value of young intensively managed pine plantations for bobwhite.

MANAGEMENT IMPLICATIONS

Intensive establishment regimes for loblolly pine plantations in our study generally provided moderate levels of winter food and loafing cover, but inadequate nesting and brood-rearing cover, for bobwhite. Chemical site preparation created a beneficial forb community but did not provide adequate bare ground for brood-rearing. Combining chemical site preparation with prescribed fire or with mechanical disturbance may be necessary to remove or incorporate debris and create conditions amenable to brood-rearing and nesting and to improve accessibility to winter forage. Management of permanent landscape features in close juxtaposition to open-canopied pine stands will likely be necessary to provide the full complement of bobwhite habitat in areas dominated by intensive pine silviculture.

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